

Physics 4240H: Assignment II

Jeremy Favro (0805980)
Trent University, Peterborough, ON, Canada

October 6, 2026

Problem 1 (6.2).

- (a) Depict using a phasor diagram the following two harmonic oscillations at point P :

$$E_1(P, t) = 2 \cos \omega t \quad \text{and} \quad E_2(P, t) = 7 \cos\left(\frac{\pi}{3} - \omega t\right).$$

- (b) Determine from this phasor approach the expression for the resultant disturbance.
(c) Confirm that the same result stems from a purely algebraic approach.

Solution 1.

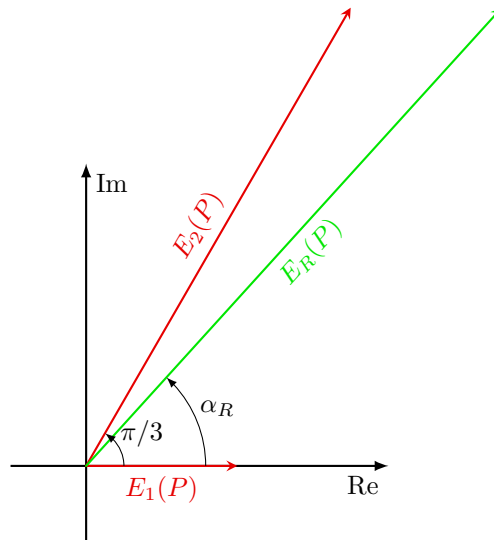
- (a) We begin by writing the fields in complex form,

$$E_1(P, t) = \text{Re}\{2e^{-i\omega t}\} \quad \text{and} \quad E_2(P, t) = \text{Re}\left\{7e^{i(\frac{\pi}{3}-\omega t)}\right\} = \text{Re}\left\{7e^{i\pi/3}e^{-i\omega t}\right\}$$

(note that change to $-\omega t$ in the first term, this makes factoring easier and doesn't affect the cosine as cosine is even) and hence their sum is

$$E_R(P, t) = \text{Re}\left\{\left(2 + 7e^{i\pi/3}\right)e^{-i\omega t}\right\}.$$

Plotting the non-oscillating (in time) part on the Argand plane,



- (b) Evidently to obtain the components of the resultant phasor we just take add the real and imaginary components of the inputs as if they were vector components. The real components are 2 and $7 \cos(\pi/3) = 7/2$ and the imaginary components are 0 and $7 \sin(\pi/3) = 7\sqrt{3}/2$ and hence the resultant phasor is

$$\left[\left(2 + \frac{7}{2}\right) + \frac{7\sqrt{3}}{2}i \right] = \frac{1}{2} [11 + 7\sqrt{3}i]$$

with angle

$$\alpha_R = \arctan(7\sqrt{3}/11) \approx 0.83$$

and magnitude $\sqrt{67}$ which gives a resultant wave of

$$E_R(P, t) \approx \operatorname{Re}\left\{\sqrt{67}e^{0.83i}e^{-i\omega t}\right\} = \sqrt{67}\cos(0.83 - \omega t).$$

(c)

$$\begin{aligned} E_R(P, t) &= 2\cos(\omega t) + 7\cos\left(\frac{\pi}{3} - \omega t\right) \\ &= \operatorname{Re}\left\{2e^{-i\omega t} + 7e^{i(\frac{\pi}{3} - \omega t)}\right\} \\ &= \operatorname{Re}\left\{[2 + 7e^{i\frac{\pi}{3}}]e^{-i\omega t}\right\} \\ &= \operatorname{Re}\{[2 + 7\cos(\pi/3) + 7i\sin(\pi/3)][\cos(\omega t) - i\sin(\omega t)]\} \\ &= \operatorname{Re}\{2\cos(\omega t) + 7\cos(\pi/3)\cos(\omega t) + 7i\sin(\pi/3)\cos(\omega t) - 2i\sin(\omega t) - 7i\cos(\pi/3)\sin(\omega t) + 7\sin(\pi/3)\sin(\omega t)\} \\ &= 2\cos(\omega t) + 7\cos(\pi/3)\cos(\omega t) + 7\sin(\pi/3)\sin(\omega t) \\ &= \frac{1}{2}\left(11\cos(\omega t) + 7\sqrt{3}\sin(\omega t)\right). \end{aligned}$$

Then we apply the identity $a\cos(x) + b\sin(x) = \sqrt{a^2 + b^2}\cos(x + \arctan(-b/a))$, where for us $a = 11$, $b = 7\sqrt{3}$, and so $\arctan(-b/a) \approx -0.83$ and $\sqrt{a^2 + b^2} = 2\sqrt{67}$. Noting that cosine is even,

$$E_R(P, t) = \frac{1}{2}\left(11\cos(\omega t) + 7\sqrt{3}\sin(\omega t)\right) \approx \sqrt{67}\cos(-0.83 + \omega t) = \sqrt{67}\cos(0.83 - \omega t)$$

indeed the same expression as with the phasor approach but with, I would say, more work.

Problem 2 (6.5). Plot and write the equation for the superposition of the following harmonic oscillations at point P :

$$E_1(P, t) = \sin\left(\frac{\pi}{18} - \omega t\right), \quad E_2(P, t) = 3\cos\left(\frac{5\pi}{9} - \omega t\right), \quad E_3(P, t) = 2\sin\left(\frac{\pi}{6} - \omega t\right)$$

where the period of each oscillation is 2 s.

Solution 2. To sum these waves algebraically we require that they all be cosines for convenience. This means we subtract $\pi/2$ from each of the sin arguments. We then rewrite the waves in complex form,

$$E_R(P, t) = \operatorname{Re}\{[\cos(4\pi/9) - i\sin(4\pi/9) + 3\cos(5\pi/9) + 3i\sin(5\pi/9) + 2\cos(\pi/3) - 2i\sin(\pi/3)]e^{-i\omega t}\}$$

This makes it (somewhat) clear that the amplitude of the resultant will be

$$E_{0R} = \sqrt{(\cos(4\pi/9) + 3\cos(5\pi/9) + 2\cos(\pi/3))^2 + (-\sin(4\pi/9) + 3\sin(5\pi/9) - 2\sin(\pi/3))^2} \approx 0.69$$

and that the angle will be

$$\alpha_R = \arctan\left(\frac{-\sin(4\pi/9) + 3\sin(5\pi/9) - 2\sin(\pi/3)}{\cos(4\pi/9) + 3\cos(5\pi/9) + 2\cos(\pi/3)}\right) \approx 0.35$$

and so the resultant wave is

$$E_R(P, t) \approx 0.69\cos(0.35 - \omega t).$$

Problem 3 (6.9). Standing waves are produced by the superposition of the wave given below and its perfect reflection from a mirror. Here, the disturbance is an electric field given in units of kV m^{-1} , while time t is in seconds and distance x is in cm:

$$E(x, t) = 7\sin\left[2\pi\left(\frac{t}{T} - \frac{2x}{\pi}\right)\right].$$

For the resultant wave, find the amplitude, wavelength, distance between nodes, velocity, and period.

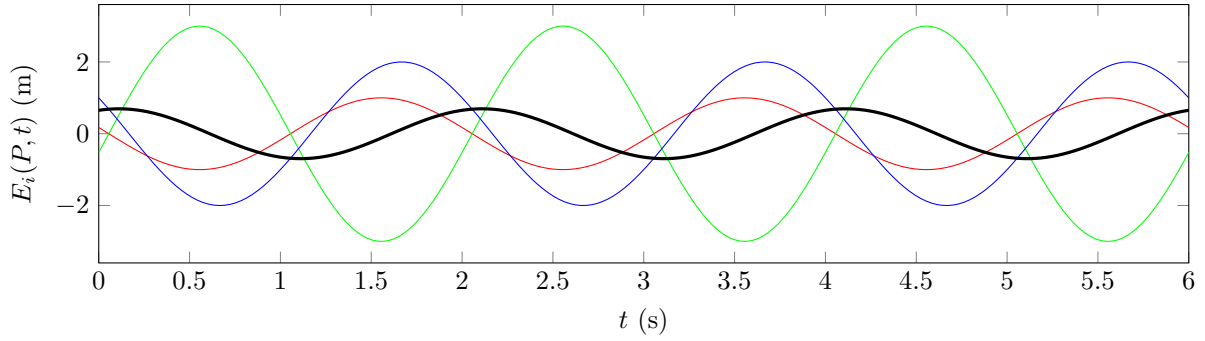


Figure 1: Red: $\sin(\frac{\pi}{18} - \omega t)$, green: $3\cos(\frac{5\pi}{9} - \omega t)$, blue: $2\sin(\frac{\pi}{6} - \omega t)$. Black is the result from summing red, green, and blue.

Solution 3. The reflected wave will be

$$E'(x, t) = 7 \sin \left[-2\pi \left(\frac{t}{T} + \frac{2x}{\pi} \right) + \pi \right].$$

The resultant wave is then

$$E_R(x, t) = 7 \left[\sin \left[2\pi \left(\frac{t}{T} - \frac{2x}{\pi} \right) \right] + \sin \left[-2\pi \left(\frac{t}{T} + \frac{2x}{\pi} \right) + \pi \right] \right]$$

which, using the identity

$$\sin(A) + \sin(B) = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$$

from the formula sheet, becomes

$$E_R(x, t) = 14 \sin \left(\frac{2\pi \left(\frac{t}{T} - \frac{2x}{\pi} \right) - 2\pi \left(\frac{t}{T} + \frac{2x}{\pi} \right) + \pi}{2} \right) \cos \left(\frac{2\pi \left(\frac{t}{T} - \frac{2x}{\pi} \right) + 2\pi \left(\frac{t}{T} + \frac{2x}{\pi} \right) - \pi}{2} \right)$$

which simplifies down to

$$E_R(x, t) = 14 \sin \left(\frac{\pi - 4x}{2} \right) \cos \left(\frac{4\pi \frac{t}{T} - \pi}{2} \right) = 14 \cos(4x) \sin \left(\frac{2\pi}{T} t \right).$$

Evidently the new wave has amplitude 14 kV m^{-1} , wavelength $\pi/2 \text{ cm}$, period T , and velocity $\pi/2T \text{ cm s}^{-1}$. Nodes occur wherever $\cos(4x) = 0$ and so a spacing of $3\pi/8$ (there are $3\pi/8 \text{ cm}$ between nodes).

Problem 4 (6.14). The *dispersive power* of glass is defined as the ratio $(n_F - n_C)/(n_D - 1)$, where C , D , and F refer to the Fraunhofer wavelengths $\lambda_{0,C} = 6563 \text{ \AA}$, $\lambda_{0,D} = 5890 \text{ \AA}$, and $\lambda_{0,F} = 4861 \text{ \AA}$. Find the approximate group velocity of a light pulse centered at the Fraunhofer D line within glass whose dispersive power is $1/30$ and for which $n_D = 1.5$.

Solution 4. Assuming Fraunhofer wavelengths are ordered sensibly (alphabetically), then we can apply

$$v_g = \frac{c}{n - \lambda \frac{dn}{d\lambda}} \bigg|_{\lambda_{\text{pk}}}$$

by approximating $dn/d\lambda \approx \Delta n/\Delta \lambda$. First $\Delta n = n_F - n_C$ which we find by rearranging $(n_F - n_C)/(n_D - 1) = 1/30$

$$(n_F - n_C)/(1.5 - 1) = \frac{1}{30} \implies \Delta n = 1/60$$

and we can find $\Delta \lambda$ to be -167.5 nm by subtracting the F and C wavelengths. Plugging these into the expression

$$v_g = \frac{c}{1.5 - 589 \text{ nm} \frac{1/60}{-167.5 \text{ nm}}} \approx 0.64c.$$

Problem 5 (IT6). A source comprising N antennas emits N waves such that the wave from the j th antenna results in an electric field at point P given by

$$E_j(P, t) = E_0 \cos(\epsilon_j - \omega t).$$

Here E_0 is the same for all antennas, and the phase difference between waves from adjacent antennas is a constant δ (so that $\epsilon_{j+1} - \epsilon_j = \delta$)

- (i) Writing the field from each antenna in the complex form $\tilde{E}_j(P, t) = E_0 e^{-i\omega t} e^{i\epsilon_j}$, where $\epsilon = \epsilon_0 + j\delta$, what is the expression for the resultant field at point P ?
- (ii) What is the expression for the amplitude of this field?
- (iii) For $N = 8$, what is the amplitude of the resultant field P when the phase difference between waves is π rad?
- (iv) For $N = 8$, what is the amplitude of the resultant field P when the phase difference between waves is 0.5 rad?

Solution 5.

- (i) Summing over all waves $E_R(P, t) = \sum_{j=1}^N E_0 e^{-i\omega t} e^{i\epsilon_j} = E_0 e^{-i\omega t} e^{i\epsilon_0} \sum_{j=1}^N e^{ij\delta}$ which is a geometric series and, using the identity (where I've changed indices since the usual sum starts at 0)

$$\sum_{j=1}^N r^{j-1} = \frac{1 - r^N}{1 - r}$$

can be rewritten as

$$E_R(P, t) = E_0 e^{-i\omega t} e^{i\epsilon_0} e^{-i\delta} \left[\frac{1 - e^{iN\delta}}{1 - e^{i\delta}} \right]$$

where the $e^{i\delta}$ accounts for the missing r^{-1} term. Looking at the possible answers, we then factor out a half (argument) exponential from top and bottom,

$$E_R(P, t) = E_0 e^{-i\omega t} e^{i\epsilon_0} e^{i\delta} e^{i(N-1)\delta/2} \left[\frac{e^{-iN\delta/2} - e^{iN\delta/2}}{e^{-i\delta/2} - e^{i\delta/2}} \right]$$

and recall that $2\text{Im}(x) = x - \bar{x}$, which is what we see in both the denominator and numerator (albeit negative and without the $2i$, but both cancel) so we end up with

$$E_R(P, t) = E_0 e^{-i\omega t} e^{i\epsilon_0} e^{i\delta} e^{i(N-1)\delta/2} \frac{\sin(\delta N/2)}{\sin(\delta/2)}$$

and $e^{i\delta} e^{i(N-1)\delta/2} = e^{i(N+1)\delta/2}$ which gives answer (D).

- (ii) Just by looking at the prefactor in the previous question, (A), $E_0 \sin(\delta N/2)/\sin(\delta/2)$.
- (iii) $E_0 \sin(4\pi)/\sin(\pi/2) = 0$, (B).
- (iv) $E_0 \sin(2)/\sin(1/4) \approx 3.68E_0$, (D).